Structure Programme C

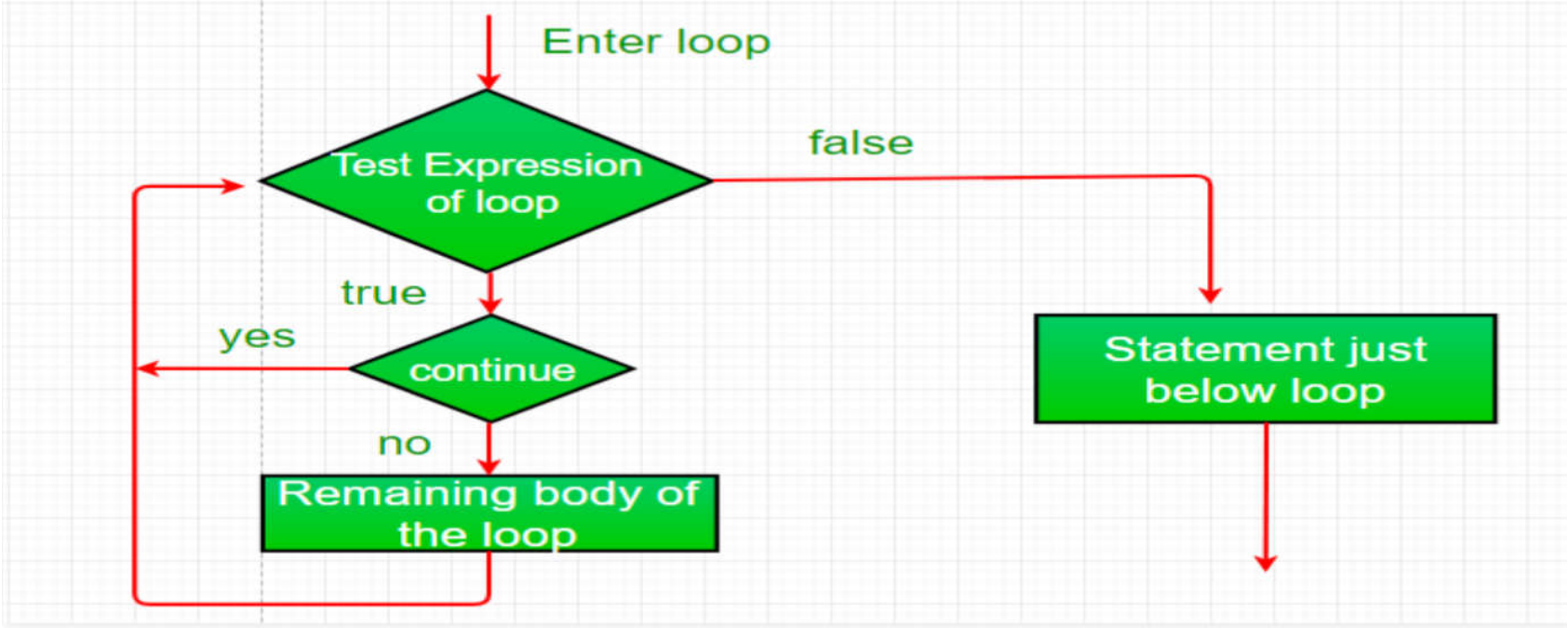
```c
#include <stdio.h> // 1-Bibliothèque et Constantes
#define Cont value

int main()
{
        // 2-Variables et Constantes


        // 3-Initialisation des Variables


        // 4-Operations


        // 5-Resultats

    return 0;
}
```

Ruptures de séquence

- **Ruptures de séquence**
  - Dans le cas où une boucle commande l'exécution d'un bloc d'instructions, il peut être intéressant de vouloir sortir de cette boucle alors que la condition de passage est encore valide.
  - Ce type d'opération est appelé une **rupture de séquence**.
  - Les **ruptures de séquence** sont utilisées lorsque des conditions multiples peuvent conditionner l'exécution d'un ensemble d'instructions.
  - Les ruptures de séquence sont réalisées par quatre instructions qui correspondent à leur niveau de travail :
    - **Ruptures de séquencement de boucle : continue ;**
    - **Ruptures de bloc: break;**
    - **Ruptures d'ordre de code : goto : Label ;**
    - **Ruptures de fonction: return value;**
    - **Ruptures de programme : exit(value) / <stdlib.h>**

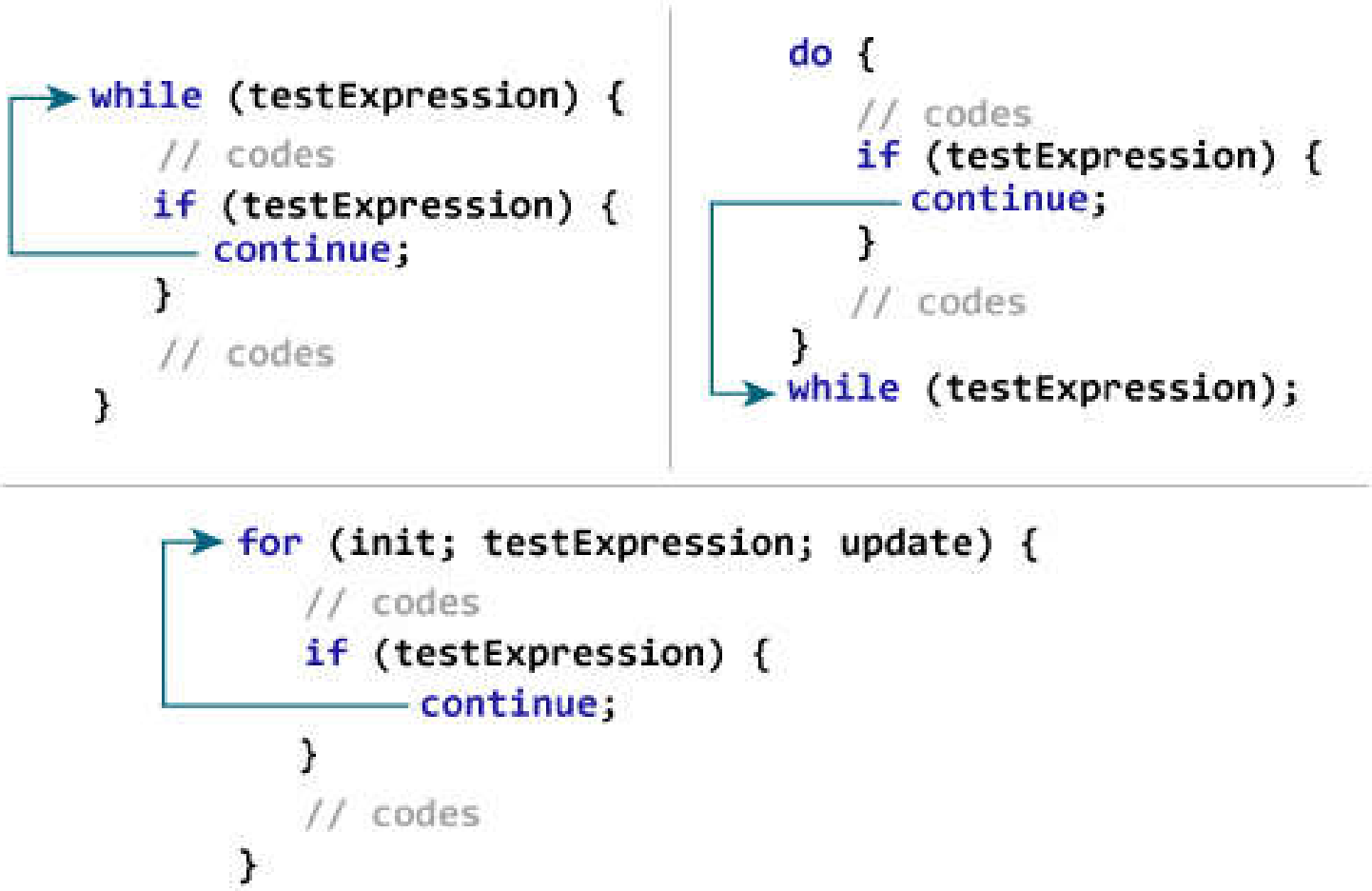
Ruptures de séquence : continue

- L'instruction **continue** est utilisée en relation avec les **boucles**.
- Elle provoque le passage à l'itération suivante de la boucle en **sautant à la fin du bloc**. Ce faisant, elle provoque la **non exécution des instructions** qui la suivent à l'intérieur du bloc.
- **Flowchart: continue**



Ruptures de séquence : continue

- **Flowchart: continue**



Ruptures de séquence : continue

- **continue**
  - **Exemple :**

```c
// Program to calculate the sum of a maximum of 10 numbers
// Negative numbers are skipped from the calculation
# include <stdio.h>
int main()
{
    int i;
    double number, sum = 0.0;

    for(i=1; i <=3; ++i)
    {
        printf("Enter a n%d: ",i);
        scanf("%lf",&number);

        if(number < 0.0)
        {
            printf("!!!!!Nombre negatif!!!!! \n");
            continue;  // = jump to next loop;
        }

        sum += number; // sum = sum + number;
    }

    printf("Sum = %.2lf",sum);

    return 0;
}
```

- **Exécution :**

```
Enter a n1: 1
Enter a n2: -2
!!!!!Nombre negatif!!!!!
Enter a n3: -3
!!!!!Nombre negatif!!!!!
Sum = 1.00
```

Ruptures de séquence : continue

- **Exercice :**

```c
int main()
{   int x=0;                          // intialisation
    while(x <10){                     // condition
        printf("%d\n",x);
        if(x==5) continue ;
          x++;                        // incrémentation
        printf("%d\n",x);
        }
    return 0;
}
```

- **Exécution :**

0 1 1 2 2 3 3 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 .....

Ruptures de séquence : continue

- **Exercice :**

```c
#include <stdio.h>

int main()
{ int x =0;
    while(x<5)
    {
    printf("%d \t",x);
    x++;
    if(x==3) continue;
    printf("%d \n",x);
    }
    return 0;
}
```

- **Exécution :**

```
0       1
1       2
2       3       4
4       5
```

Ruptures de séquence : continue

- **Exercice :**

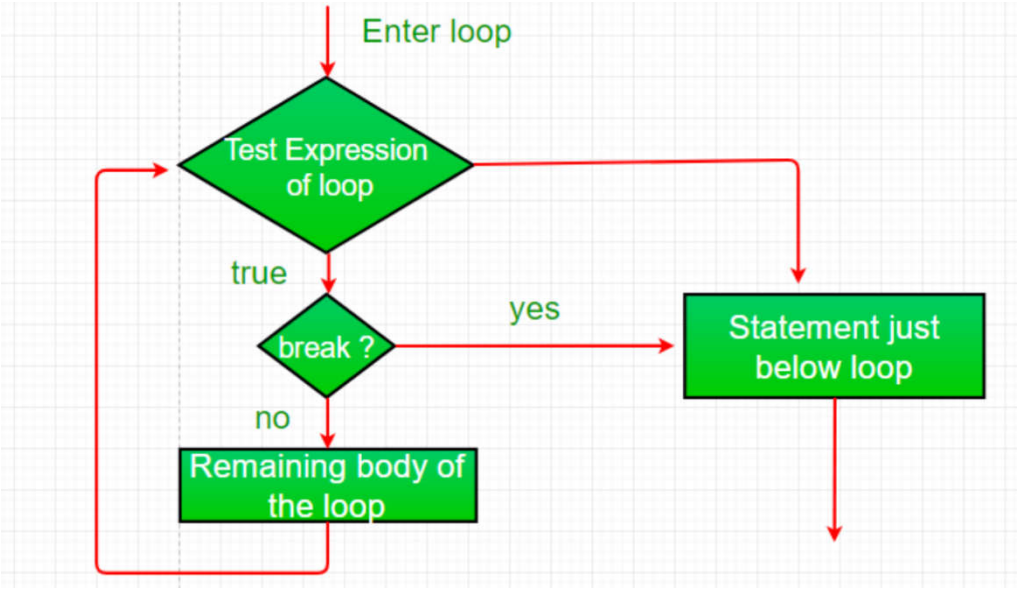
```c
#include <stdio.h>

int main()
{ int x =0;
    while(x<5)
    {
    printf("%d \t",x);
    x++;
    if(x>3) continue;
    printf("%d \n",x);
    }
    return 0;
}
```

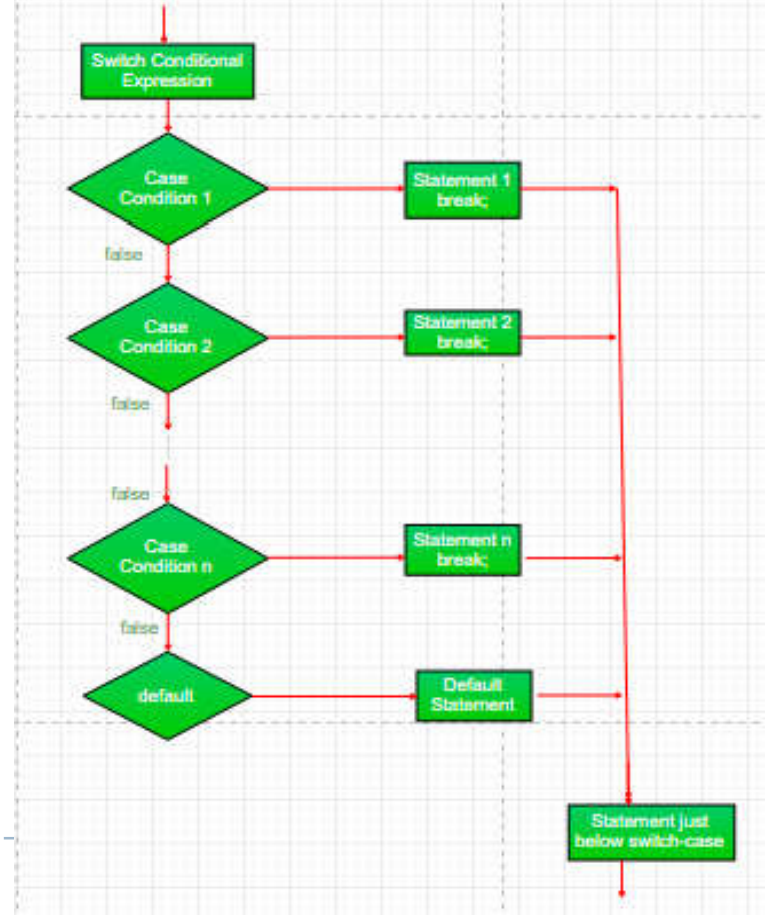
- **Exécution :**

```
0       1
1       2
2       3
3       4
```

Ruptures de séquence : break

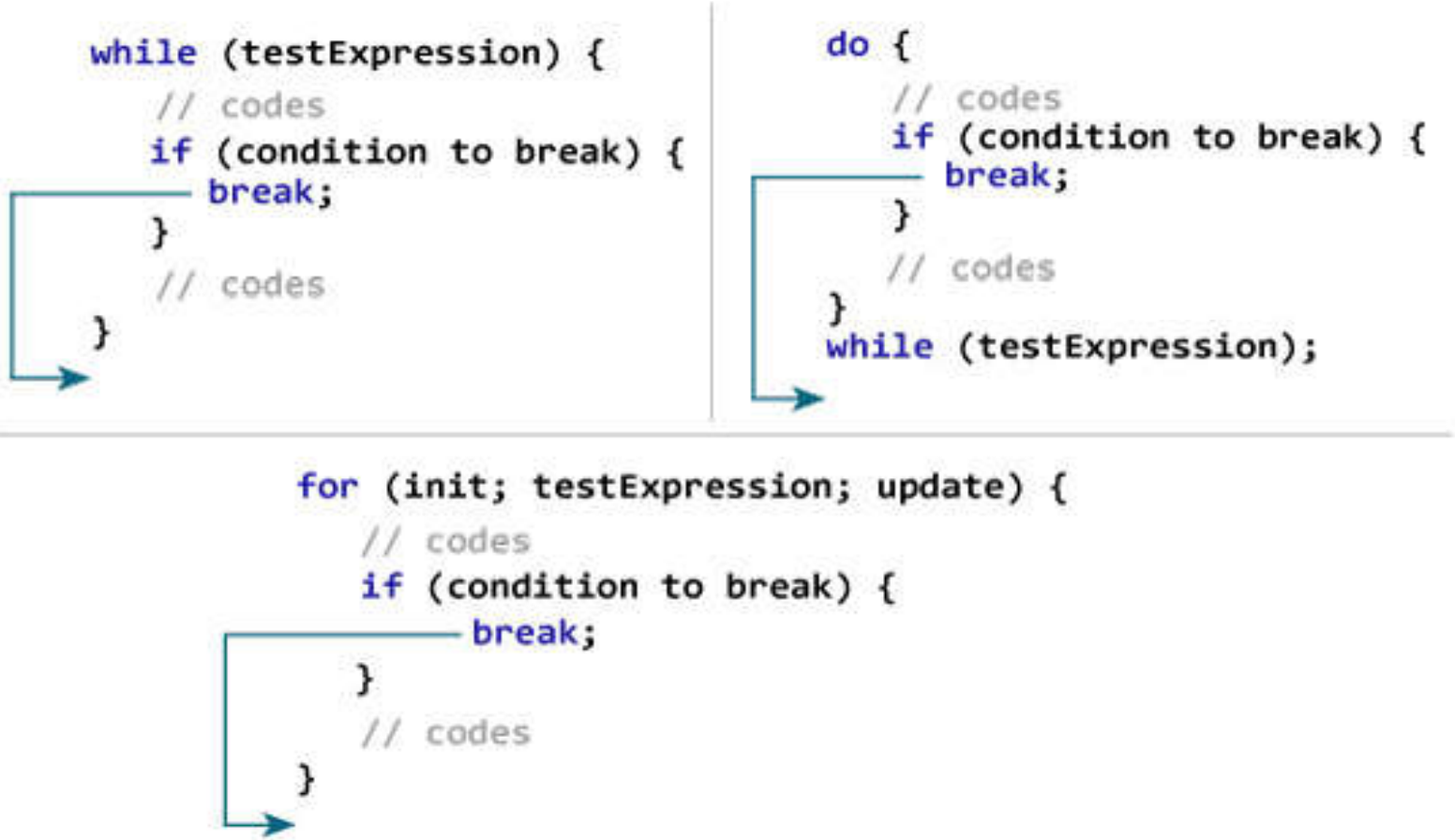
- L'instruction **break** est utilisé déjà dans le **switch**.
- L'instruction **break** permet de sortir d'un bloc d'instruction associé à une instruction **répétitive** ou **alternative** contrôlée par les instructions switch , for, un while ou un do while.
- **Flowchart: break continue**





Ruptures de séquence : continue

- **Flowchart: break**



Ruptures de séquence : break

- **break**
  - **Exemple : Boucle**

```c
// Program to calculate the sum of a maximum of 10 numbers
// If a negative number is entered, the loop terminates
# include <stdio.h>
int main()
{
    int i;
    double number, sum = 0.0;

    for(i=1; i <=4; ++i)
    {
        printf("Enter a n%d: ",i);
        scanf("%lf",&number);

        // If the user enters a negative number, the loop ends
        if(number < 0.0)
        {
            printf("!!!!Arret de Boucle!!!! \n");
            break;
        }

        sum += number; // sum = sum + number;
    }

    printf("Sum = %.2lf",sum);

    return 0;
}
```

- **Exécution :**

```
Enter a n1: 1
Enter a n2: 2
Enter a n3: -4
!!!!Arret de Boucle!!!!
Sum = 3.00
```

Ruptures de séquence : break

- **break**
  - **Exemple : Switch**

```c
# include <stdio.h>
int main()
{
    int i;
    printf("Enter a i: ");
    scanf("%d",&i);
        switch (i)
        {
        case 0:
            printf("i is zero.");
            break;
        case 1:
            printf("i is one.");
            break;
        case 2:
            printf("i is two.");
            break;
        default:
            printf("i is greater than 2.");
        }
    return 0;
}
```

- **Exécution :**

```
Enter a i: 0
i is zero.
```

```
Enter a i: 1
i is one.
```

```
Enter a i: 6
i is greater than 2.
```

Ruptures de séquence : break

- **Exercice :**

```c
int main()
{
  int n;

while (-1)
{
    printf("Tapez un nombre Positif ! ");
    scanf("%d", &n);
    if(n>0) break;
}

 printf(" Suite du Programme  ");
    return 0;
}
```

- **Exécution :**

Ruptures de séquence : break

- **Exercice :**

```c
int main()
{
  int n=0;

while (n<5)
{
    printf("nombre %d ",n);
    if(n==3) break;
}

 printf(" Suite du Programme  ");
    return 0;
}
```

- **Exécution :**

Ruptures de séquence : break

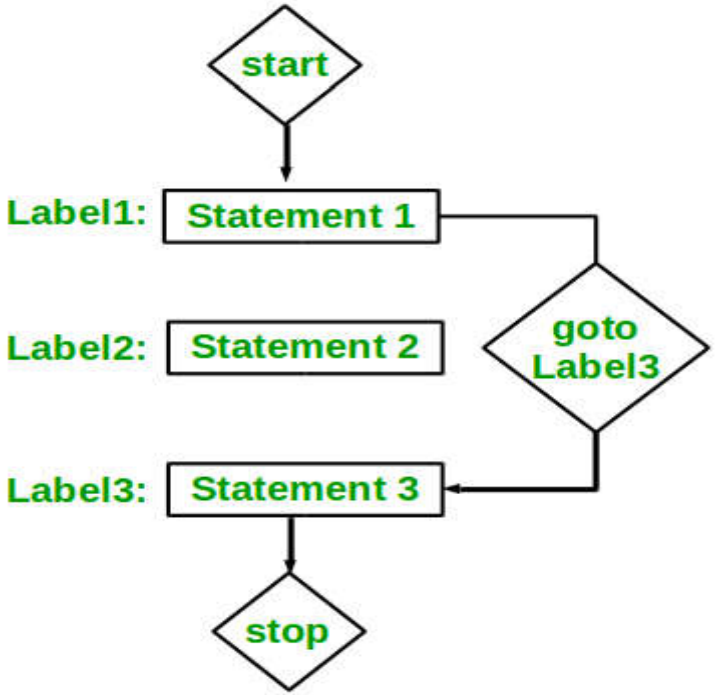
- **Exercice :**

```c
int main()
{
  int n=0;

while (n<5)
{
    printf("%d ",n);
    n++;
    if(n==3) break;
    printf("%d ",n);
}
    return 0;
}
```

- **Exécution :**

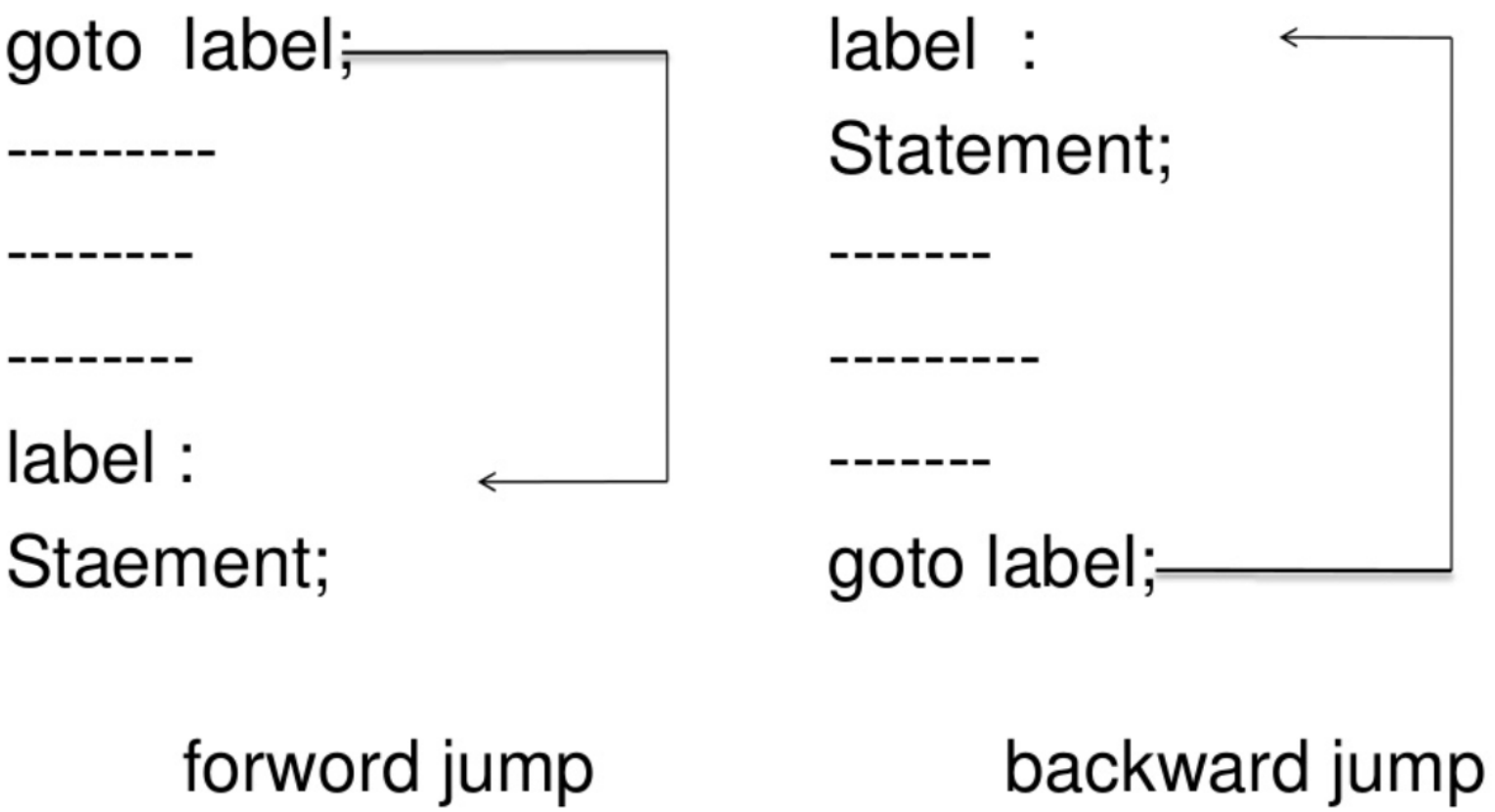
Ruptures de séquence : goto

- Dans la programmation C, *goto* est utilisé pour modifier la séquence normale de l'exécution du programme en transférant le contrôle à une autre partie du programme
- **Flowchart**: **goto**



Ruptures de séquence : goto

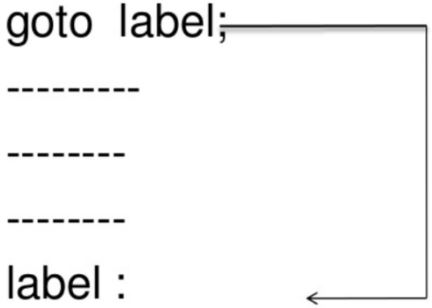
- **Syntaxe**: **goto**



Ruptures de séquence : goto

- **Syntaxe: goto Forword jump**

```c
#include <stdio.h>
int main()
{   int num;
    printf("Enter number \n");
    scanf("%d", &num);
    if (num > 0)
        printf("Number= %d \n",num);
    else
        // jump to war
        goto war;

    // some instructions

war:
    printf("End of program");

    return 0;
}
```



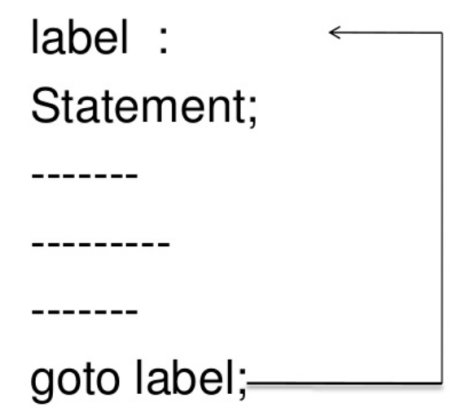
forword jump

```
Enter number
10
Number= 10
End of program
```

```
Enter number
-10
End of program
```

Ruptures de séquence : goto

- **Syntaxe: goto Backward jump**




```c
#include <stdio.h>
int main()
{    int n, i=0;

   printf("Enter a number: ");
   scanf("%d",&n);

   start:
   printf("%d\t",i);
   i++;
   if(i<n) goto start;

    return 0;
}
```

```
Enter a number: 6
0       1       2       3       4       5
```

Ruptures de séquence : goto

- **Syntaxe: goto Backward jump**

```
label :
Statement;
-------
---------
-------
goto label;
```

backward jump

```c
#include <stdio.h>
int main()
{    int n;

   start:
   printf("Enter a number: ");
   scanf("%d",&n);

   if(n<0) goto start;

   printf("Suite du programme ");

    return 0;
}
```

```
Enter a number: -2
Enter a number: -4
Enter a number: -5
Enter a number: 10
Suite du programme
```

Ruptures de séquence : goto

- goto
  - Exemple :

```c
#include <stdio.h>
int main()
{
  int Totalmarks;

  printf(" \n Please Enter your Subject Marks \n ");
  scanf("%d", & Totalmarks);

  if(Totalmarks >= 50)
   {
     goto Pass;
   }
  else
     goto Fail;

  Pass:
     printf(" \n Congratulation! You made it \n");

  Fail:
     printf(" \n Better Luck Next Time \n");

 return 0;
}
```

- Exécution :

```
Please Enter your Subject Marks
30

Better Luck Next Time
```

```
Please Enter your Subject Marks
88

Congratulation! You made it

Better Luck Next Time
```

Ruptures de séquence : goto

- **goto**
  - **Exemple : return value;**

```c
#include <stdio.h>
int main()
{
  int Totalmarks;

  printf(" \n Please Enter your Subject Marks \n ");
  scanf("%d", & Totalmarks);

  if(Totalmarks >= 50)
   {
     goto Pass;
   }
  else
     goto Fail;

  Pass:
     printf(" \n Congratulation! You made it \n"); return 1;

  Fail:
     printf(" \n Better Luck Next Time \n");

 return 0;
}
```

- **Exécution :**

```
Please Enter your Subject Marks
90

Congratulation! You made it
```

Ruptures de séquence : goto

- **goto**
  - **Exemple : return value;**

```c
#include <stdio.h>
int main()
{
  int Totalmarks;

  printf(" \n Please Enter your Subject Marks \n ");
  scanf("%d", & Totalmarks);

  if(Totalmarks >= 50)
   {
     goto Pass;
   }
  else
     goto Fail;

  Pass:
     printf(" \n Congratulation! You made it \n"); return 1;

  Fail:
     printf(" \n Better Luck Next Time \n");

 return 0;
}
```

- **Exécution :**

```
Please Enter your Subject Marks
90

Congratulation! You made it
```

Ruptures de séquence : goto

- **goto**
  - **Exemple : exit(value)**

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
  int Totalmarks;

  printf(" \n Please Enter your Subject Marks \n ");
  scanf("%d", & Totalmarks);

  if(Totalmarks >= 50)
   {
     goto Pass;
   }
  else
     goto Fail;

  Pass:
     printf(" \n Congratulation! You made it \n"); exit(1);

  Fail:
     printf(" \n Better Luck Next Time \n");

 return 0;
}
```

- **Exécution :**

```
Please Enter your Subject Marks
88

Congratulation! You made it
```

Control statements in C